

The Run In

Six days to the Micromart demonstration — Wednesday 2 September 2026

Three repositories, six systems and two relays already stand up. This is the audit of what each screen actually reads, the twelve things between here and a two-hour demonstration that never once shows a modelled number, and the day-by-day plan to close them.

Written **Friday 28 August 2026** Demonstration **Wed 2 Sept** Working days remaining **4**
Evidence **verified on live systems**

01 Where we stand, according to the machine

run 28 Aug 2026

Nothing below is recalled from a document. Every row was produced by running the check this afternoon against the deployed estate and Micromart's production server.

CHECK	VERDICT	WHAT IT PROVES
test:relay	● PASSED	95,409 tracked loans read through the relay. SQL 703 ms, round trip 1,411 ms. DATETIME survived the wire as a Date . Writes refused at the socket.
test:crb:prod	● 8 / 8	Every Metropol service entitled against production :22225, egressing from the whitelisted address. The E003 blocker is gone.
test:live	● 14 LIVE · 0 DEGRADED	All six subdomains answer 200. One sign-in carries across every origin on the .servicesuitecloud.com cookie. No screen fell back to a degraded state.
test:desk	● PASSED	Eleven checks on the collections floor, including a shadow write composed, stored and not executed. Pipeline and mirror both armed false.
Group book read	● 673 MS	Micromart's four entities aggregated in one round trip through the relay — 161,862 borrowers, 327,709 loans, KES 562.9m outstanding.
Interchange filters	● 2 OF 4 MEMBERS	Bloom filters published for Micromart Fintech (79 tokens) and Axe Boresha (14,149). Micromart Africa and Axe Stawi are declared but absent.
Fintech book read	● HTTP 503	"The relay host has no connection string for Micromart Fintech." One missing variable on salesmaster, described in F1.

THE HEADLINE

The hard problems are behind us. A serverless deployment now reads a database that has no internet route, a bureau answers from a registered address, and one sign-in opens six front doors. What remains is not infrastructure — it is that **three of the six systems are still reading the wrong database.**

Every screen in the suite resolves to one of two data paths: **live** — `runReadOnlyQuery()` through the relay into Micromart's SQL Server — or **our own Postgres**, which for a bridged lender holds almost nothing. The split below was traced from each page's imports to its query, not read off a roadmap.

SYSTEM	SCREENS READING LIVE	WHERE THE NUMBERS COME FROM
ConnectDesk	12 / 12 	CollectBox joined to Serviceconnect in one query plan. The floor, the queues, the case file, the timeline, the pipeline. Shadow queue reads our own records by design — that is what it is for.
PeopleHub	3 / 3 	The roster from UserMaster, officers joined through Borrowers.EntityAgent to the floor and what it recovered.
Ledgerly	3 / 3 	The real journal, paged with an account filter; disbursement and collection read from two databases nobody has reconciled.
Lending Console	3 / 27 	Borrower list and field worklist read through live. The dashboard overlays live KPIs but leaves four panels modelled . Everything else — applications, loans, collections, repayments — is Postgres.
Analytics Studio	1 / 15 	Only the group roll-up reads live. The other fourteen aggregate in Postgres via lib/analytics/engine.ts — a 199-loan book .
Micro Eazy (PWA)	16 / 24 	Core banking screens run on Micromart's own live API. The ecosystem screens go through /api/portal/* — and "My loan" refuses bridged lenders outright .
Estate	38 / 84 	45% of the estate reads Micromart. The plan takes it to 88% by Tuesday.

Live — read through the relay Modelled or our own Postgres

The Analytics Studio is the screen the Data & Analytics manager opens, and it is the one that reads least. Both figures below are Micromart's loan book. They differ by a factor of sixteen hundred.

LIVE, THROUGH THE RELAY

327,709

loans across four entities · 161,862 borrowers · KES 562.9m outstanding. This is what ConnectDesk, PeopleHub and Ledgerly read today.

IN OUR POSTGRES

199

loans · 162 borrowers · 9 branches. This is what the Analytics Studio and most of the Lending Console read today.

This is not a defect that crept in — it is the deliberate design working as specified. A bridged lender's customers are *never* mirrored into our database, so the console can never show a stale copy of somebody else's book. The borrower list honours that by reading through. The studio was written before the bridge existed and still aggregates locally, so it honours the same rule by finding almost nothing.

ENTITY	NAME	BORROWERS	LOANS	ACTIVE	OLB (KES)	ARREARS (KES)	LAST LOAN
3002	Micromart Africa Limited	142,393	264,178	10,122	91,084,484	28,121,968	28 Aug
3003	Micromart Check off	2,448	3,664	437	470,890,580	115,837,161	28 Aug
3005	Micromart Fintech — the portal lender	17,021	59,867	93	915,261	101,861	27 Aug
3004	Micromart IPF — empty	0	0	0	0	0	—
Group		161,862	327,709	10,652	562,890,325	144,060,990	

BLOCKERS — ONE VARIABLE AND ONE COLUMN STAND BETWEEN US AND THE EXCHANGE

F1 The relay cannot read Micromart Fintech ● BLOCKER

`relay-host/.env.example` never mentions `MICROMART_FINTECH`, and `warmPools()` in `sql-relay.ts` warms only micromart and axe. Both Interchange members on that server — 3005 and 3002 — resolve through the micromart-fintech org, so a single absent variable takes out half the exchange.

```
$ npm run interchange:members
Micromart (services) unreachable - SQL relay returned 503.
"The relay host has no connection string for Micromart Fintech."
```

F2 Half the Interchange membership is unpublished ● BLOCKER

`NODE_MEMBERS` declares four. The deployed registry serves two: Fintech (79 tokens) and Axe Boresha (14,149), both built 26 August. Missing is KE/LENDER/3002 — Micromart Africa, 142,393 borrowers, the anchor member the whole exposure argument rests on. Consequence of F1; fixed by re-running the publish once the variable is set.

F3 The group roll-up crashes on Axe's server ● BLOCKER

`getGroupBook()` selects `onboardingLatitude / onboardingLongitude`. Axe's Serviceconnect predates those columns, so the one analytics surface that *does* read live throws on the second-largest book in the ecosystem. Guard the two terms with `COL_LENGTH()`.

```
AXE - ERR Invalid column name 'onboardingLongitude'.
```

LIVE-DATA GAPS — THE REASON THIS DOCUMENT EXISTS

F4 The Analytics Studio aggregates the wrong database ● MAJOR

Fourteen of fifteen surfaces call `headline`, `timeSeries`, `cube` or `cohorts` in `lib/analytics/engine.ts`, which groups in Postgres through `orgTx`. For Micromart that is a 199-loan book, so the board view will greet the room with *"There is no open book in this cut."* The studio's entire data surface is those six functions — which is also why it is tractable in a weekend.

F5 Four console dashboard panels stay modelled forever ● MAJOR

`MODELLED_PANELS = ["series", "aging", "productMix", "branches"]`. `applyLive()` overlays real KPIs and then sets `simulated: false` — true of the headline figures, false of the time series, the aging buckets, the product mix and the branch table, which are still produced by a seeded PRNG. The screen labels each region honestly, which is the right behaviour and still not a thing to project.

F6

The borrower's own loan screen refuses bridged lenders ● MAJOR

`/api/portal/my-loan` returns `bridged: true` and the message *"Your Micromart loan is managed on the lender's own system."* A real Micromart borrower opening the ecosystem's My-loan screen is deflected. Same for `/api/portal/ladder`, which counts Postgres rows. Both live equivalents are already written and tested — `getCustomer360()` and `checkGraduation()` — so this is wiring, not new SQL.

F7

Console operations screens read Postgres ● ACCEPT

Applications, loans, collections, disbursements, repayments, products and workflows all resolve locally. This is *correct* — they are origination surfaces for a book we originate, and Micromart originate in their own core. It is nonetheless 24 screens that must not be opened on a Micromart session during the demonstration. Drive origination from Mular Credit instead, where the book is genuinely ours.

F8

Interchange scoring and learning run on seeded holdings ● ACCEPT

`seed-holdings.ts` and `seed-ledger.ts` build the overlapping population, the bitemporal ledger, the decision history and the declined applicants that reject inference recovers. The exposure engine itself queries the two live members. Say the distinction out loud on the Score and Learning screens rather than letting somebody find it.

HYGIENE — SMALL, CHEAP, AND EACH ONE IS VISIBLE FROM THE ROOM

F9

The artwork check reports a failure that is not one ● COSMETIC

`verify-live.ts` probes six `.png` masters; `artwork.ts` serves `.webp`. PeopleHub's master is a `.jpg`, so the PNG 404s and the run reports "5/6 artworks". All six doors serve their plate — confirmed 200, 9,884 bytes. Point the check at the file the app reads.

F10

The relay-host bundle is untracked ● RISK

Eleven files — the host-verification script, the service installer, both relays and the failure taxonomy — exist only on this laptop. They are the deployment that makes the cloud work, and the document that explains what to do when it stops. Commit them.

F11

The ecosystem registry says two systems are not deployed ● COSMETIC

`registry.json` carries `live: false` for the Interchange and the PWA, and a note that `portal.servicesuitecloud.com` has no DNS record. Both now answer 200 with their real titles. The registry is the file every script trusts; a stale one is how a fresh machine gets the wrong idea.

F12

Founding-cohort books are not reachable from the relay ● POST-DEMO

Ten qualifying entities on the `.198` server — NJB, Buy Simu, ATICO and seven more, ~43,000 borrowers between them — need `SERVICESUITE_CONN_BUYSIMU` on the relay host to join the exchange. Not demo-critical, and the single highest-value thing to do the week after.

Ordered so that the cheapest unblocks land first and the one genuinely large piece — the Analytics Studio — gets the weekend it needs. Tuesday is rehearsal, not development.

TONIGHT • FRI 28 Unblock the exchange

≈ 3 h

- | | | |
|-------|--|------|
| A1 | Set <code>MICROMART_FINTECH</code> on <code>salesmaster</code> . Same connection string as <code>SERVICESUITE_CONN_MICROMART</code> . Add it to <code>.env.example</code> , add the slug to <code>warmPools()</code> , restart <code>MicroEazySqlRelay</code> . Verify with <code>npm run interchange:members</code> . | 30 m |
| <hr/> | | |
| A2 | Publish the two missing members. <code>npm run interchange:publish -- --all</code> . Expect ≈10,100 tokens for 3002 and ≈115 for Axe Stawi. Confirm four filters at <code>/api/filters</code> . | 1 h |
| <hr/> | | |
| A3 | Guard the two <code>onboarding*</code> columns in <code>getGroupBook()</code> so the roll-up survives Axe's older schema. | 1 h |
| <hr/> | | |
| A4 | Commit <code>relay-host/</code> to the meta-repo, land the two documentation edits, and bump the submodule pointers. | 30 m |

SAT 29 – SUN 30 The Analytics Studio, on live data

≈ 14 h

Build `lib/analytics/engine-live.ts` answering the same six signatures against Serviceconnect through the relay, and branch on `ctx.bridged`, which `studioContext` already resolves. The measure definitions are not guesswork — `analytics/group.ts` already reproduces Micromart's own `MainDashboard` arithmetic, including the 90-day non-performing rule, verified against their live dashboard.

- | | | |
|-------|---|-----|
| B1 | <code>headline</code> + <code>timeSeries</code> — unlocks the board view, portfolio, collections, cashflow, funnel, risk and borrowers. Seven of fifteen surfaces. | 6 h |
| <hr/> | | |
| B2 | <code>cube</code> by branch, product, officer, region, channel and risk band — unlocks agents, branches, products, regions, channels. Five more. | 5 h |
| <hr/> | | |
| B3 | <code>filterOptions</code> + <code>inceptionDate</code> — the filter bar's real axes, and "since inception" meaning the first loan on <i>this</i> book. | 2 h |
| <hr/> | | |
| B4 | <code>cohorts</code> + the Explorer. Cut line. Both degrade honestly today; ship them after the demo if Saturday runs long. | 3 h |

THE ALTERNATIVE, AND WHY NOT

The other route is a nightly mirror of Micromart into Postgres. It is rejected. It makes the studio a *stale copy* of a book the borrower list next door reads live, which is the exact contradiction a sharp manager will find; it puts 327,709 loans and 1.95 million schedule rows into a database that is not ours to hold; and it is an ETL project, not four days' work. Read through, as everything else in the suite already does.

MON 31 The console dashboard and the borrower's phone

≈ 7 h

- | | | |
|-------|--|-----|
| C1 | Feed the four modelled panels from B1 and B2. Then remove the <code>simulate()</code> fallback on the bridged path so the dashboard degrades honestly instead of reaching for a cinematic fiction. | 4 h |
| <hr/> | | |
| D1 | Wire <code>/api/portal/my-loan</code> to <code>getCustomer360()</code> and <code>/api/portal/ladder</code> to <code>checkGraduation()</code> for bridged orgs. A Micromart borrower then opens the app and sees their real loan. | 3 h |

TUE 1 SEPT Rehearsal and freeze

all day

- | | | |
|-------|---|-------|
| E1 | The full sweep — <code>test:relay</code> , <code>test:crb:prod</code> , <code>test:desk</code> , <code>test:collectbox</code> , <code>test:live</code> , <code>test:access</code> , <code>interchange:verify</code> . Every one green before anything else happens. | 1 h |
| <hr/> | | |
| E2 | Point the artwork check at <code>.webp</code> (F9) and refresh <code>registry.json</code> (F11). | 30 m |
| <hr/> | | |
| E3 | Walk the two hours with a timer, on the deployed site, over the public internet, signed in the way a supervisor signs in. Every route compiles on first render — warm all of them. | 2.5 h |
| <hr/> | | |
| E4 | Freeze. No deploys after the rehearsal passes. Confirm <code>SQL_RELAY_ALLOW_WRITES</code> unset and both funnel mappings listed. | 30 m |

IF THE WEEKEND SLIPS

B4 goes first — the Explorer and cohorts degrade honestly and nobody asks for them unprompted. B2 goes second, which costs five ranking surfaces but keeps the board view, portfolio and collections live. **B1 is not negotiable:** without it the Analytics act opens on an empty book, and that is the one moment in two hours where the room would stop believing the rest.

Built so the argument compounds. The floor first, because it is undeniable and it is theirs. The borrower's phone in the middle, because it turns a lender's tool into a market. The exchange last, because it only means something once the room has seen five systems read one book.

CLOCK	MIN	ACT	THE BEAT, AND THE LINE THAT CARRIES IT
0:00	8	The topology /suite, cold	Open the launcher and say nothing for twenty seconds. Six tiles, each carrying a number read when the page rendered. Then the relay in one diagram: their database has no internet route, so we moved the socket, not the application. "Every figure on this page was read from your SQL Server when it loaded. Nothing here is seeded, cached or remembered."
0:08	22	ConnectDesk /desk → /queue → a case → /shadow	The floor as it stands this minute. Their seven queues, their commission bands, their agents ranked by cash rather than dials. Sort by longest untouched — the sort that finds the book going quietly cold. Open a case: every loan, every call, every payment, merged from five sources. Pick a disposition, show the "This will" block, submit, then show the composed SQL sitting unexecuted in the shadow queue. "Everything ConnectDesk would write to your database, in full, held for you to read first. Arming it is one environment variable."
0:30	16	Lending Console /console → borrowers → field	The dashboard against their own MainDashboard figures — same OLB, same PQS, same 90-day rule, because we reproduced their arithmetic rather than inventing ours. Then the borrower list: 17,021 customers, paged out of their server, never copied. Finish on the field worklist — the borrowers with no location, which is an operations problem nobody has been able to see.
0:46	15	Analytics Studio board → portfolio → collections → group	The board view in the order a board asks: how big, is it going bad, is money coming in, who brings it, what sells. Every figure compared against the same elapsed days last period. Close on the group roll-up. "You run four entities on one server and you have never seen them side by side, because your own dashboard scopes to whoever signed in."
1:01	13	PeopleHub & Ledgerly /people/officers → /books/journal	Officers sorted by weakest coverage — the thirteen with a book on the floor who took no payment in thirty days. Then the journal: three years and 6.4 million postings that nothing has ever read, each row linking to its case on the floor. "This needed your roster, your ledger and your collections floor in one query. No system in this building can currently produce it."
1:14	17	Micro Eazy the phone, on the projector	Install it from the home screen. Verify a phone, see a real loan, run the M-Pesa statement crunch, pull a live Metropol report — 8 of 8 services entitled, from a whitelisted address — then Ratiba and an STK prompt landing on the handset. Finish on the limit ladder and the decision explanation. "Same book, same second, borrower's side."
1:31	19	The Interchange consent → exposure → log → governance	The consent gate refusing a query with no reference. Then an exposure check across four member books, answered without any member learning who asked, and without the registry ever holding a national ID. Show the hash-chained log and try to tamper with it. Name the seeded surfaces before anyone asks. "Nothing is pooled. Identity is destroyed at the edge. What crosses is a token and four numbers."
1:50	10	The bridge /desk/pipeline	The close, and it is one number. 95,409 loans on their collections floor; not one of them is entity 3005. Walk the six sections as the argument they are laid out to be — the gap, the mechanism, the 98% agreement with their own nightly ageing job, the queues, the

CLOCK	MIN	ACT	THE BEAT, AND THE LINE THAT CARRIES IT
			commission, and the customers who would arrive tomorrow. Press Preview . Statements compose. Nothing runs. "We did not invent an ageing rule. We reproduced yours and measured the agreement."
2:00	120	Questions	Budget nothing. If the room is engaged the questions have already been arriving since 0:30, and the acts above absorb them.

07

Wednesday morning

before the room fills

On salesmaster

```
Get-Service MicroEazySqlRelay, MicroEazyCrbRelay  
tailscale funnel status # BOTH 443 and :8443
```

A bare `tailscale funnel 8787` takes 443 and silently unpublishes the CRB relay. Bureau calls then fail the quiet way — dropped at Metropol's edge, no `api_code`, nothing in a log.

- **Do it during working hours.** The floor's numbers reset at midnight EAT. At 09:00 you see a day building; at 00:30 you see almost nothing.
- **Say the write posture out loud early.** `SQL_RELAY_ALLOW_WRITES` unset, `COLLECTBOX_POSTING_ENABLED` unset, pipeline and mirror both false. It is a strength, not a caveat, and saying it first stops it being asked as a suspicion later.
- **If the amber banner appears** it now means exactly one thing: the relay is not answering. Host asleep, off the tailnet, or the funnel dropped. `npm run test:relay` separates the four cases by name.
- **Afterwards** — `npx tsx scripts/setup-desk-demo.ts --suspend`. Those three sign-ins open a real lender's book.

From anywhere

```
npm run test:relay # ALL CHECKS PASSED  
npm run test:crb:prod # 8/8 entitled  
npm run test:live # 14 live, 0 degraded  
npm run interchange:members # 4 filters
```

Then open all six subdomains once. First render of each route compiles; after that everything is 1.5–4s.

The suite's credibility rests on every screen degrading honestly, and that only works if the person presenting does the same. Four things to name from the front.

- **PeopleHub has three empty tables and says so.** Payroll, leave and contracts. We will not invent a payroll figure — and the reason to say that out loud is that it is the same promise that makes the balances trustworthy.
- **Their data stops in places.** Promises end November 2024, tasks August 2025, call detail records September 2023. These are findings about their process, not gaps in ours. Show them; they are worth money to the room.
- **The Interchange's scoring and learning screens run on a seeded population.** The exposure engine queries live member books; the model training history does not exist yet because the exchange is six days old. Name the boundary on the screen it applies to.
- **Nine indexes are proposed and none applied.** `loanSchedule` (1.95m rows), `Loans` (338k), `CallLogs` (1.34m) and `PayedAmount` (1.15m) carry no indexes at all. It is their database, so it is their decision — and their own reporting is almost certainly paying the same tax.